

# Imesh Ranaweera

☎ +1 352 682 9931 | ✉ imeshchamara57@gmail.com | ✉ imesh.ranaweera@ufl.edu | 📍 United States

🌐 [imesh-chamara](#) | 🔄 [ImeCMR](#) | 🌐 [Website](#)

## PROFILE

PhD student in Chemistry at the University of Florida, specializing in theoretical and computational chemistry. Research interests span computational methods for drug discovery, machine learning-guided molecular property prediction, and the application of artificial intelligence to chemical and biological systems. Currently contributing to the Molecular Dynamics Data Bank America (MDDBa) project the first MDDB node outside Europe and in the United States and actively engaged in developing and benchmarking machine-learned force fields for biomolecular simulations.

## RESEARCH INTERESTS

**Keywords:** Computational Chemistry · Drug Development · Machine Learning · Artificial Intelligence · Molecular Dynamics · Force Field Development

## EDUCATION

### University of Florida, College of Liberal Arts and Sciences

2024 – 2029 (Expected)

*Doctor of Philosophy (Reading) in Chemistry*

Gainesville, FL, USA

- **GPA:** 4.00/4.00
- **Coursework:** Chemical Bonding and Spectra, Computational Chemistry, Statistical Thermodynamics, Biological System Modeling, AI in Biology, AI in Agricultural and Life Sciences

### University of Colombo, Faculty of Science

2019 – 2023

*Bachelor of Science (Honors) in Computational Chemistry*

Colombo, Sri Lanka

- **GPA:** 3.70/4.00 | **Class:** First Class Honors
- **Chemistry Coursework:** Computational Chemistry, Computational Programming in Chemistry, Advanced Molecular Modeling, Physical Chemistry, Biochemistry, Organic Chemistry, Inorganic Chemistry, Spectroscopy, Organometallic Chemistry
- **Computing/Data Coursework:** Machine Learning and Neural Computing, Data Structures and Algorithms, Database Systems, Fundamentals of Software Engineering, Data Analytics, Computational Biology, Logic Programming

### Ananda College

2014 – 2017

*G.C.E. Advanced and Ordinary Level Examinations*

Colombo, Sri Lanka

- **A/L (2017):** Biology – A, Chemistry – B, Physics – B | Z Score: 1.6644
- **O/L (2014):** Nine Distinctions

## RESEARCH EXPERIENCE

### Molecular Dynamics Data Bank America (MDDBa)

2024 – Present

*Graduate Research, Molecular Recognition Group (Perez Lab)*

University of Florida, USA

- Contributing to the establishment of MDDBa, the **first MDDB node outside Europe** and the first in the United States, as part of the Perez Lab at the University of Florida

- Involved in data curation, infrastructure development, and interoperability efforts to expand global access to molecular dynamics simulation data

## Undergraduate Research Project

2022 – 2023

Supervisor: Prof. Samantha Weerasinghe, Dept. of Chemistry

University of Colombo, Sri Lanka

- Title:** *Investigation of the effects of selected hydrotropes on the solubility of drug-like molecules from Sri Lankan flora: a computational approach*
- Investigated the hydrotropic activity of pyrogallol and resorcinol on selected bioactive drug-like molecules sourced from Sri Lankan flora, with the aim of characterizing their solubility-enhancing mechanisms at the molecular level
- Performed extensive molecular dynamics (MD) simulations using **GROMACS** with the **KBFF20 force field** at 300 K and 1 bar to model aqueous hydrotrope–solute interactions
- Authored custom **Fortran 95** programs for automated system preparation, cluster separation analysis, and free-molecule concentration calculations, enabling efficient large-scale MD data processing
- Demonstrated that **pyrogallol exhibits stronger hydrotropic effects** than resorcinol for the selected molecules, providing molecular-level insight into hydrotrope selectivity relevant to pharmaceutical formulation

## PUBLICATIONS

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### Journal Article

2026

- Imesh Ranaweera, Alberto Perez. *Beyond Classical Force Fields: Physics-Driven Assessment of the Grappa Machine-Learned Force Field on the FoldBind Dataset*. *ChemPhysChem*, 2025.  
<https://doi.org/10.1002/cphc.202500815>

### Conference Abstract

2023

- I.C.M. Ranaweera, S. Weerasinghe. *Investigation of the effects of selected hydrotropes on the solubility of drug-like molecules from Sri Lankan flora: a computational approach*. 79th Annual Scientific Sessions, Sri Lanka Association for the Advancement of Science (SLAAS).

## TEACHING & ACADEMIC EXPERIENCE

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### Graduate Research Assistant

May 2026 – Present

Department of Chemistry, Molecular Recognition Group (Perez Lab)

University of Florida, USA

- Conducting computational chemistry research focused on molecular dynamics, machine-learned force fields, and drug discovery applications

### Graduate Teaching Assistant

Aug. 2024 – May 2026

Department of Chemistry

University of Florida, USA

- Assisted faculty with lecture delivery and course coordination for undergraduate chemistry courses
- Conducted weekly discussion and recitation sessions for undergraduate students
- Held regular office hours to provide individualized academic support
- Proctored examinations and evaluated assignments, problem sets, and laboratory reports

### Teaching Assistant (Visiting)

2023 – 2024

Department of Chemistry

CINEC Campus, Sri Lanka

- Assisted in delivering a Certificate Course in Computational Chemistry: Computer Aided Drug Design

## Teaching Assistant

Aug. 2023 – Jul. 2024

*Dept. of Computation and Intelligent Systems, UCSC*

University of Colombo, Sri Lanka

- Facilitated computer laboratory practical sessions and tutorials for first- and second-year undergraduates
- Co-supervised second- and third-year group projects, providing technical and research guidance
- Evaluated assignments, tutorials, and practical assessments across multiple modules
- **Subjects:** Introduction to Computing, Data Structures and Algorithms, Programming and Problem Solving, Rapid Application Development, Middleware Architecture
- Organizing Committee Member, **23rd ICTer International Conference**
- Organizing Committee Member, **I Am a Researcher Competition 2023**

## AWARDS, CERTIFICATIONS & ACHIEVEMENTS

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### Achievements

#### Global Winning Team Member, BioML Challenge (Bits to Binders 2024–2025)

2025

- Member of PDNALab, the global winning team of the BioML Challenge: Bits to Binders competition

#### HiPerGator Pioneering Team Award

2025

- Team member recognized with the HiPerGator Pioneering Team Award at the University of Florida

#### Champions – Inter-University Chemistry Magic Competition

2022

- Led team to first-place finish in the inter-university chemistry magic competition

### Certifications

Research Security Training (Combined), *CITI Program*

2026

Research Security Training, *CITI Program*

2026

Physical and Natural Sciences Responsible Conduct of Research, *CITI Program*

2026

Fundamentals of Deep Learning, *NVIDIA*

2026

Supervised Machine Learning: Regression and Classification, *DeepLearning.AI*

2024

Introduction to Artificial Intelligence, *IBM*

2023

Drug Discovery, *University of California San Diego*

2023

Introduction to Quantum Computing, *Udemy*

2023

Observation of Near-Earth Objects and Main Belt Asteroids via Pan-STARRS Image Analysis 2020, 2021, 2022

## PRESENTATIONS & ACADEMIC ACTIVITIES

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**Oral:** *Beyond Classical Force Fields: Physics-Driven Assessment of the Grappa Machine-Learned Force Field on the FoldBind Dataset*, University of Florida, USA

**Oral:** *Investigation of the effects of selected hydrotropes on the solubility of drug-like molecules from Sri Lankan flora: a computational approach*, University of Florida, USA

**Oral:** *Biochemistry of Human Hair*, University of Colombo, Sri Lanka

**Interview:** Team interview of Prof. Rodolphe Barrangou for CH 4002 Seminar and Essay, University of Colombo, Sri Lanka

**Project:** Fortran 95 program to simulate a model salt in liquid argon for CH 3032 Computational Programming in Chemistry, University of Colombo, Sri Lanka

## SKILLS

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**Programming :** Python, Fortran, SQL

**Scientific Software :** AMBER, GROMACS, MELD, AlphaFold, BioEmu & aSAM, MDANCE, Gaussian 09W, GaussView 5.0, RasMol, RasWin, PyMOL, VMD, ChimeraX, Xmgrace, HyperChem Professional, WinMopac 7.21, ChemDraw, BURAI, ORCA

**AI & LLM Tools :** Claude Code / Codex / Opencode

**Laboratory :** Practical computational chemistry; qualitative and quantitative analysis of inorganic and organic compounds; basic synthetic organic chemistry techniques; basic characterization techniques; fundamental physical chemistry experiments

**Instrumentation :** Centrifuge machine, UV-Vis Spectrometer, pH meter, Water bath sonicator

**Operating Systems :** Linux, macOS, Windows

**Languages :** English, Sinhala

## LEADERSHIP & SERVICE

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**Vice President,** Chemical Society, University of Colombo 2022–2023

**Member,** Leo Club, University of Colombo 2019–2023

**Member,** Astronomical Society, University of Colombo 2019–2023

**Senior Member,** Science Union of Ananda College 2015–2017

**Senior Member,** Biological Society of Ananda College 2015–2017

**Member,** Saukyadana Unit, Ananda College 2011–2013

## PROFESSIONAL MEMBERSHIPS

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**American Chemical Society (ACS)**

**Sri Lanka Association for the Advancement of Science (SLAAS)**

## ENGLISH PROFICIENCY

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**IELTS** (2022-08-26): Overall **6.5** | Listening: 6.5, Reading: 6.5, Writing: 6.0, Speaking: 7.0

## SPORTS & OTHER ACTIVITIES

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Cricket, Hockey, Swimming, Badminton

## REFERENCES

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**Dr. Alberto Perez**

Associate Professor and William R. Kenan, Jr. Term Professor; Associate Director, Quantum Theory Project  
Department of Chemistry, University of Florida, USA

✉ perez@chem.ufl.edu